Research Report: SMA120 Trend Filter for BTC (Long-Only)

Prepared 20 October 2026. Data: Binance BTCUSDT spot (daily), 2017-08-17 to 2026-09-29. This document is a research summary based on backtests, not financial advice.

1. Summary

The rule tested is simple: hold BTC when the close is above the 120-day moving average (SMA120), otherwise hold cash. The rule cuts maximum drawdown by roughly half and passed a single out-of-sample (OOS) test with criteria locked before the results were seen. However, its return is far lower than the in-sample figures, and the CAGR advantage over buy&hold; almost disappears in the OOS. The main benefit is risk reduction, not additional return.

The numbers to rely on are the OOS figures (Calmar around 1.1, CAGR around 30%), not the in-sample figures.

Metric	Buy&hold; OOS (2024-26)	SMA120 OOS (2024-26)	SMA120 in-sample (2018-23)
CAGR	28.1%	31.9%	75.5%
Max drawdown	-53.0%	-28.4%	-32.3%
Calmar (CAGR / |MDD|)	0.53	1.12	2.34
Time in market	100%	58%	52%

2. Background

This research grew out of a series of searches for an edge in crypto perpetual futures that did not produce a viable strategy:

- Long/short Donchian breakout (N=100): discovery Sharpe 1.29 fell to -0.30 in validation, and all key criteria failed.
- Cross-sectional behavior exploration (60 IC hypotheses): many patterns were statistically significant, but 0 of 36 long-short portfolios passed after realistic costs and funding.

From there the goal was simplified: capture BTC market returns while suppressing bear market drawdowns, rather than hunting for alpha. The approach is a long-only trend filter.

3. Strategy rules

Component	Specification
Instrument	BTC/USDT spot (long-only, no leverage)
Signal	Daily close > average of the last 120 closes (including today) → hold BTC; otherwise cash
Execution	Signal at the close of day t, position effective from the open of day t+1 (no lookahead)
Costs	0.10% per side (fee + slippage); stress at 0.20% and 0.30%
Cash return	0% (conservative assumption)
Calculation	Compounding, open-to-open returns
Check frequency	Once a day after the 00:00 UTC close

4. Data and methodology

- **Data:** daily BTCUSDT spot klines from Binance Vision, 3,332 days with no missing days. The in-sample reproduction matches earlier results (CAGR 75.5%, MDD -32.3%, Calmar 2.34).
- **In-sample period:** 2018-06-13 to 2023-12-30 (300-day warm-up so all lookbacks are valid). It covers the 2018 bear, the Covid crash, the May 2021 crash, and the 2022 bear.
- **OOS period:** 2024-01-01 to 2026-09-29 (1,003 days). This data was untouched during the research and was run only once.
- **Search space:** 65 configurations in the initial screening, then a robustness map of 108 configurations (SMA 80-160 in steps of 10, times 12 filter variants).
- **Choosing L:** uses smoothed Calmar (the average of three adjacent L values), not the raw peak, to avoid narrow spikes.
- **Main metrics:** CAGR, MDD, Calmar, comparison with a static allocation at the same exposure (a control to check that timing really adds value), and up/down capture.

OOS criteria locked before the run: (1) Calmar > buy&hold;, (2) Calmar > static allocation with the same exposure, (3) |MDD| ≤ 70% of buy&hold; MDD, (4) CAGR > 0 at fee 0.20%, (5) at least 4 of 5 neighbors (L=100, 110, 120, 130, 140) pass criteria 1 and 3.

5. In-sample results (2018-06-13 to 2023-12-30)

5.1 SMA120 vs buy&hold;

Metric	Buy&hold;	SMA120
CAGR	39.9%	75.4%
Max drawdown	-76.6%	-32.3%
Calmar	0.52	2.34
Time in market	100%	52%
Position changes per year	0	6.7
Calmar at fee 0.20% / 0.30%	n/a	2.28 / 2.22

5.2 Returns during crash episodes

Episode	Buy&hold;	SMA120
2018 H2	-51%	-11.7%
Covid crash 2020	-51%	-22.3%
May 2021 crash	-53%	-22.2%
2022 bear	-75%	-27.7%

Calmar by sample half: SMA120 3.49 and 1.61, versus buy&hold; 1.25 and 0.17 (before and after 2021-01-01). The advantage appears in both halves.

5.3 Parameter map (plain SMA)

L	80	90	100	110	120	130	140	150	160
Calmar	1.06	1.00	1.62	2.16	2.34	1.96	1.65	1.32	1.13
MDD %	-52.5	-50.5	-38.5	-34.0	-32.3	-35.0	-40.4	-44.8	-45.8

All L values in the 80-160 range beat buy&hold; (Calmar 0.52). The peak is around 110-130, and in the initial screening SMA200-300 was much weaker (SMA200 Calmar 0.63, SMA300 0.51, equal to buy&hold;). So the "classic SMA200" is not automatically better.

5.4 Do additional filters help?

Each filter was tested against the plain SMA (plain SMA smoothed Calmar = 2.15). A filter counts as adding value if it improves the score by at least 0.15 and is not weaker in both sample halves.

Variant	Smoothed Calmar difference vs plain SMA	Adds value?
+momentum 60 days	-0.16	No
+momentum 90 days	-0.67	No
+exit on volatility spike (80th percentile)	-0.68	No
+exit on volatility spike (90th percentile)	-0.02	No
+hysteresis 3%	-0.43	No
+hysteresis 5%	-0.52	No
Gradual position (momentum 60)	-0.06	No
Gradual position (momentum 90)	-0.35	No
+volatility scaling 60%	-0.17	No
+volatility scaling 80%	-0.02	No
Weekly check	-1.05	No

Conclusion: no filter adds value, so the plain version was chosen for simplicity. Among other families tested in the early stage, volatility targeting and drawdown-from-ATH exposure were not useful, and the SMA ensemble (CAGR 50.8%, MDD -51.3%, Calmar 0.99) was no better than SMA120.

6. OOS results (2024-01-01 to 2026-09-29)

6.1 Locked criteria

#	Criterion	Result	Value
1	Calmar > buy&hold;	PASS	1.12 vs 0.53
2	Calmar > static allocation (same exposure)	PASS	1.12 vs 0.55
3	|MDD| ≤ 70% of buy&hold; MDD	PASS	-28.4% vs limit -37.1%
4	CAGR > 0 at fee 0.20%	PASS	+30.2%
5	Robust: ≥ 4 of 5 neighbors pass	PASS	5 of 5

Verdict: PASSED all five criteria.

6.2 SMA neighbors (plateau check)

L	CAGR	MDD	Calmar	CAGR fee 0.20%	CAGR fee 0.30%
100	26.8%	-36.1%	0.74	24.8%	22.8%
110	34.2%	-28.9%	1.18	32.4%	30.6%
120	31.9%	-28.4%	1.12	30.2%	28.5%
130	31.0%	-25.6%	1.21	29.4%	27.8%
140	27.2%	-27.3%	1.00	25.7%	24.2%

SMA120 is not the highest in the OOS (SMA130 and SMA110 are slightly better). That is a sign of a plateau, not a narrow peak. At fee 0.30%, SMA120 CAGR (28.5%) is almost the same as buy&hold; (28.1%), so the return advantage is very sensitive to costs. The MDD advantage is not.

6.3 By calendar year

Year	SMA120	BTC	SMA120 MDD	BTC MDD
2024	+92.3%	+121.1%	-25.5%	-26.2%
2025	+4.2%	-6.3%	-19.1%	-32.0%
2026 (272 days)	+6.7%	-4.5%	-13.2%	-39.5%

In 2024 (a bull market) the strategy lagged buy&hold; by about 29 points. That is the price of drawdown insurance.

6.4 Largest BTC drawdown in the OOS

Period	Return
BTC buy&hold; 2025-10-06 to 2026-06-30	-53.0%
SMA120 over the same window	-22.0%

SMA120 exited at around 114,000 (28 October 2025) and re-entered at around 77,000 (18 April 2026). The latest position has been open since 2026-08-20 (open around 69,335).

6.5 Trade pattern

Group	Count	Range
Winning trades	4	+0.8% to +54.1% (three largest: +54.1%, +46.4%, +16.0%)
Small losing trades	13	-0.4% to -6.5%
Total completed	17	
Position changes per year	12.7	(in-sample: 6.7)

Profits are concentrated in three trades, while 13 other trades are small losses (whipsaw). Position turnover in the OOS is almost double the in-sample level.

7. Interpretation: why Calmar fell from 2.34 to 1.12

The drop in Calmar comes from CAGR, not MDD. Drawdown protection held up and was even slightly better.

	In-sample	OOS	Change
Buy&hold; CAGR	39.9%	28.1%	-11.8 pts
SMA120 CAGR	75.5%	31.9%	-43.6 pts
SMA120 minus buy&hold;	+35.6	+3.8	-31.8 pts
SMA120 MDD	-32.3%	-28.4%	improved

The weaker market (2025-26 bear) explains only about a quarter of the CAGR decline. The rest is the shrinking advantage over buy&hold.;

The most likely causes (hypotheses, not provable from this data): the very strong 2020-21 bull trend favored a trend strategy, and SMA120 was picked as the best of many configurations, so the in-sample figures are optimistic. More whipsaw in the OOS also eroded returns.

Therefore, plan with the OOS numbers, not the in-sample ones.

8. Risks and limitations

Risk	Explanation
Selection bias	SMA120 was chosen from more than 100 configurations on the same data. In-sample figures are optimistic.
Thin evidence	Only about two bear markets in-sample and one large drawdown in the OOS. Passing means consistent, not proven.
OOS no longer clean	After the first run, the 2024+ OOS has been used. Re-tuning would repeat overfitting.
Lags in bull markets	In 2024 the gap was about -29 points. Without a bear market, this strategy loses to buy&hold.;
Whipsaw	13 of 17 OOS trades were small losses. Discipline is needed to sit through consecutive losses.
Cost sensitivity	At fee 0.30% per side, the CAGR advantage disappears. The MDD advantage remains.
Execution assumptions	Daily open-to-open. Real slippage in volatile markets, exchange outages, and taxes are not modeled.
Cash risk	Parking cash in stablecoins or on an exchange carries depeg and custody risk. Cash return is assumed to be 0%.
Reference price	BTC/USDT on Binance, not USD. The spread is usually small but can widen under stress.
MDD expectation	It is reasonable to expect an MDD of about -30% to -45%, not -20%.

9. Practical use

- Check once a day after the 00:00 UTC close: compare the BTCUSDT close with the average of the last 120 closes (including today).
- Close above SMA120 → hold BTC. Close below → hold cash. Execute at the next day's open.
- Expect roughly 6 entry-exit cycles per year, with most losses small in sideways markets.
- Start with small capital and decide in advance where the cash will be parked.
- Realistic expectations from the OOS: CAGR around 30%, MDD around -25% to -45%, Calmar around 1.0-1.2. In a bull market without a bear phase, results will be below buy&hold.;

10. Next steps

1. Do not re-tune L or the filters based on the OOS that has already been used.
2. Run a true forward test: record daily signals starting now (paper trading or small capital). Data after 2026-09-29 is the only OOS that remains clean.
3. Compare periodically with buy&hold; to see whether the drawdown protection still works.
4. Optional: add data from other assets with longer histories to test whether the same trend filter produces similar patterns.

Appendix: files and settings

File	Purpose
`download_btc_spot.py`	Download daily BTCUSDT spot klines and convert to CSV with headers (2025+ microsecond timestamps normalized)
`explore_behavior.py`	Loader and cross-sectional behavior analysis (used as a library)
`btc_mdd_reduction.py`	Comparison of 65 configurations of MDD-reduction methods
`btc_sma_robust.py`	SMA 80-160 robustness map with 12 filter variants, costs, and validation by half and episode
`btc_oos_sma120.py`	Single OOS test with locked criteria and a lock file

Main settings: fee 0.10% per side (stress 0.20% and 0.30%), 300-day warm-up, cash return 0%, compounding, execution at open t+1.